

Principal Computational Scientist
The Answer Project (TAP Sciences)
Adjunct Professor of Data Science
Southern Methodist University
Dallas, Texas

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Professional Summary

Computational scientist with a Ph.D. in theoretical and computational chemistry and more than a decade of experience spanning computational drug design, molecular simulation, scientific machine learning, and GPU-accelerated high-performance computing (HPC). Leads the selection and application of computational methods, datasets, and AI/ML models used to generate and prioritize drug-development hypotheses, and translates those methods into scalable, production-grade discovery workflows. Recognized for impactful interdisciplinary work and awarded for innovative computational strategies.

Core Competencies

- Computational Drug Design, Structure-Based Molecular Modeling, and Cheminformatics
- Molecular Simulation, Binding Free Energies, and Physics-Based Property Prediction
- Scientific Machine Learning, Graph and Generative Models, and Surrogate Modeling
- Quantum Chemistry and Electronic-Structure Method Development
- End-to-End *In Silico* Discovery Workflows on GPU-Accelerated and Distributed HPC
- Scientific and Technical Leadership, Project Scoping, and Partnership Development

Scientific and Technical Expertise

Molecular Simulation	CHARMM, NAMD, LAMMPS, Amber, GROMACS, OpenMM; binding free energies, reaction paths, potential energy surfaces, and spectroscopic properties
Quantum Chemistry	Gaussian, CFOUR, Q-Chem, GAMESS, PySCF, DFTB+, Molpro, NWChem, ORCA, GPAW; classical MM/MD, DFTB, DFT, MP, and CC methods
Machine Learning	PyTorch, PyTorch Geometric (PyG), GPyTorch, scikit-learn, TensorFlow, RAPIDS
Languages	Python, C, C++, Fortran, Rust, Julia, R
Parallelism	MPI, CUDA, OpenMP/OpenACC, NCCL, (Open NV)SHMEM, KOKKOS, RAJA
Platforms and Tools	Linux, Git, Slurm, Kubernetes, Docker, Apptainer/Singularity, Enroot, Nix

Experience

Principal Computational Scientist — May 2025 to Present

[The Answer Project \(TAP Sciences\)](#)

Remote

- Lead the selection and application of computational methods, datasets, AI/ML approaches, and molecular simulations used to generate and prioritize drug-development hypotheses.
- Develop structure-based molecular modeling, machine learning-driven property and activity prediction, and end-to-end *in silico* workflows supporting drug discovery programs, including oncology.
- Direct computational research initiatives and funding opportunities in pharmacological and materials discovery with U.S. federal agencies, national laboratories, major research centers, and industry.
- Design the HPC architecture and scientific software strategy underpinning chemical discovery and simulation programs.

Adjunct Professor of Data Science — January 2021 to Present

[Department of Statistics and Data Science](#)

Southern Methodist University, Dallas, Texas

- Designed and taught graduate-level high-performance computing, machine learning, and data science courses.

Principal Scientist — May 2024 to May 2025

[O'Donnell Data Science and Research Computing Institute](#)

Southern Methodist University, Dallas, Texas

- Directed cross-disciplinary strategic initiatives and funding opportunities, translating research needs into technical programs and computing strategies.
- Developed and scaled applications and workflows for chemical simulation and data-intensive scientific machine learning.

HPC Applications Scientist — May 2015 to May 2024

Office of Information Technology: Research and Data Science Services & Center for Research Computing

Southern Methodist University, Dallas, Texas

- Advised interdisciplinary research teams on computational methods, profiling, parallelization, GPU acceleration, and HPC adoption; optimized and scaled scientific codes and workflows across successive HPC clusters.
- Instrumental in the procurement, integration, and optimization of the ManeFrame II, NVIDIA DGX SuperPOD, and M3 clusters.
- Designed and led HPC and data science courses, workshops, and training sessions for advanced users and researchers.
- SMU President's Award for Innovation (2020) for contributions to computational strategies in the COVID-19 campus response.

Postdoctoral Fellow — January 2014 to May 2015

Department of Chemistry

Southern Methodist University, Dallas, Texas

- Developed and applied the Rigid Residue Scan method through extensive molecular dynamics simulations to identify key allosteric residues and characterize residue entropic roles in protein allostery.

Education

Doctor of Philosophy in Chemistry, Theoretical and Computational Chemistry — August 2009 to May 2014

Southern Methodist University, Dallas, Texas

Description of the Strength of Chemical Bonds Utilizing Local Vibrational Modes

Master of Science in Chemistry, Theoretical and Computational Chemistry — August 2007 to May 2009

University of Texas at Dallas, Dallas, Texas

Area Per Ligand as a Function of Nanoparticle Radius: A Theoretical and Computer Simulation Approach

Bachelor of Science in Chemical Engineering, Minors in Mathematics and Chemistry — August 2001 to May 2006

Texas Tech University, Lubbock, Texas

Prairie Grass Ethanol Production Pilot Plant Facility and Optimization

Project Management Certificate — December 2023

Southern Methodist University, Dallas, Texas

Selected Publications

- R. Kalescky, H. Zhou, J. Liu, and P. Tao, "Rigid Residue Scan Simulations Systematically Reveal Residue Entropic Roles in Protein Allostery," *PLoS Computational Biology* **12**(4), 1–21 (2016). doi:[10.1371/journal.pcbi.1004893](https://doi.org/10.1371/journal.pcbi.1004893)
- R. Kalescky, J. Liu, and P. Tao, "Identifying Key Residues for Protein Allostery through Rigid Residue Scan," *The Journal of Physical Chemistry A* **119**(9), 1689–1700 (2015). doi:[10.1021/jp5083455](https://doi.org/10.1021/jp5083455)
- R. Kalescky, E. Kraka, and D. Cremer, "Accurate Determination of the Binding Energy of the Formic Acid Dimer: The Importance of Geometry Relaxation," *The Journal of Chemical Physics* **140**(8) (2014). doi:[10.1063/1.4866696](https://doi.org/10.1063/1.4866696)
- R. Kalescky, E. Kraka, and D. Cremer, "Identification of the Strongest Bonds in Chemistry," *The Journal of Physical Chemistry A* **117**(36), 8981–8995 (2013). doi:[10.1021/jp406200w](https://doi.org/10.1021/jp406200w)

Full publication list: [ORCID 0000-0002-1416-4044](https://orcid.org/0000-0002-1416-4044).